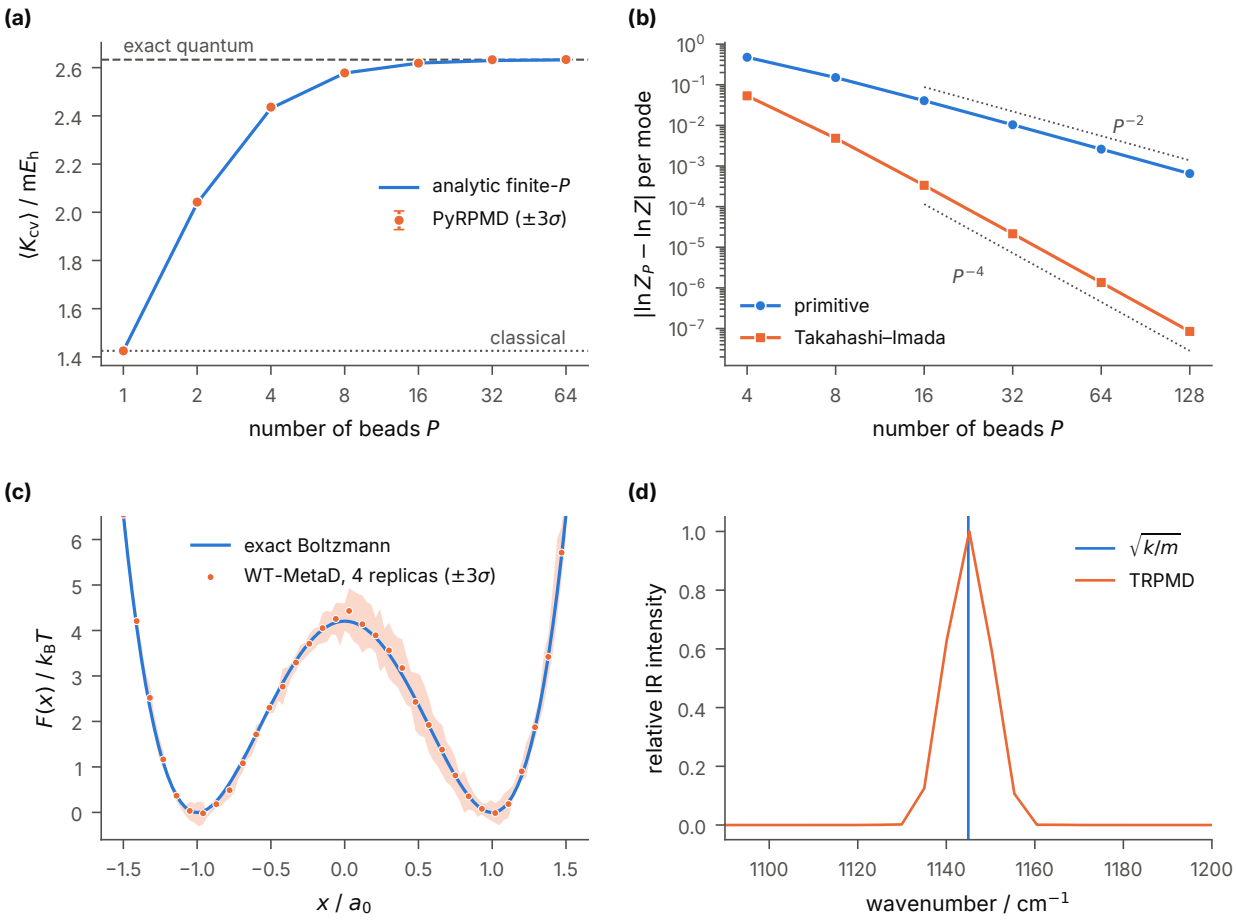
(a)
exact quantum
classical
analytic finite-P
PyRPMD (±3σ)
⟨K_cv⟩ / mE_h
number of beads P
(b)
|ln Z_P − ln Z| per mode
P^−2
P^−4
primitive
Takahashi–Imada
number of beads P
(c)
exact Boltzmann
WT-MetaD, 4 replicas (±3σ)
F(x) / k_B T
x / a_0
(d)
√(k/m)
TRPMD
relative IR intensity
wavenumber / cm^−1